

Syllabus for CIS 241

System-level Programming and Utilities

Instructor & Meeting Informations

Instructor: Anthony Langley

Pronouns: he/him

Office: MAK-C-2-310

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Meeting times:

- Section 03 - M/W/F 12:00-12:50pm
- Section 04 - M/W 4:30-5:45pm

Meeting location: Mackinac Hall A-1-105 (EOS Lab, section 03) & Online for Section 04

Personal Student Drop-in Hours (time set aside to help YOU!):

- Mondays 1:00 pm -2:00 pm in the Computing Success Center (MAK-A-1-101)
- Wednesdays: 1:00 pm – 2:00 pm @ Office
- MWF: 11:00 am – 12:00 pm @ Office
- And by appointment (Zoom available)

Note: this syllabus was modified from Dr. Erin Carrier's CIS 241 syllabus and Dr. Alexander Lalejini's syllabus.

Course Information

Course Objectives

After taking this course, students will be able to:

- Use Unix-like command-line interfaces to perform basic system operations, including on remote machines
- Understand and manipulate staple features of command-line environments
- Write meta-programs in a shell or scripting language (e.g., bash)
- Employ the fundamental concepts of git for version control
- Write, compile, and debug simple C programs
- Demonstrate dynamic memory allocation and pointer usage in C

Textbook

This class has no required textbooks.

If you want supplemental info, you may find one or more of the books listed below useful. Links are provided to the library's website, where **all books can be accessed online *for free***.

- Linux
 - [Linux Pocket Guide, 3rd edition - Barret \(2016\)](#)
 - [The Linux Command Line, 2nd edition - Shotts \(2009\)](#)
 - [Linux in a Nutshell, 6th edition - Siever, Figgins, Love, and Robbins \(2009\)](#)
- Programming in C
 - [Understanding and Using C Pointers, 1st edition - Reese \(2013\)](#)
 - [C in a Nutshell, 2nd edition - Prinz and Crawford \(2015\)](#)
 - [Head First C, 1st edition - Griffiths and Griffiths \(2012\)](#)
 - [C Programming Absolute Beginner's Guide, 3rd edition - Miller and Perry \(2013\)](#)

Prerequisites

Must have completed CIS 163 or CIS 260

Grading

Graded Assignment Breakdown

Your final grade will depend on your grades on these activities:

- Check-in assignments (estimated 11): 25%
- Projects (estimated 4): 30%
- Basic command line proficiency quiz: 5%
- Midterm Exam: 20%
- Final Exam: 20%

Grade Scale

Your final letter grade will be assigned as follows.

Note: I will round up automatically (ceiling function). For example, a 93.2% will become a 94%.

A	94%	C	74%
A-	90%	C-	70%
B+	87%	D+	67%
B	84%	D	60%
B-	80%	F	Below 60%
C+	77%		

Class technology

Blackboard Ultra

Link: <https://lms.gvsu.edu>

Blackboard will be used for two purposes: 1) to hold the official gradebook and 2) as an initial landmark to direct you to the other course technologies.

PrairieLearn

Link: <https://us.prairielearn.com/>

PrairieLearn will be used to handle both projects and (most) check-in assignments in this course. We will discuss how to enroll in the course on PrairieLearn soon. PrairieLearn is free to you as a student.

Most assignments will be auto-graded on PrairieLearn. *However*, make sure you check grades on Blackboard once they are transferred over. That is where final grades will be maintained.

Discord

Link: <https://discord.com/>

I personally find Blackboard clunky for class-wide communication, so I use Discord to distribute information and facilitate discussions.

The main idea of having a shared communication platform is to facilitate discussion between students. If you have a general question about the class, please ask it there! Additionally, I strongly encourage you to answer each other's questions! I will jump in when needed and when I am available.

If you have a private question meant only for me (e.g., discussing accommodations, a specific code question), **please email me instead**.

Finally, while Discord is generally a lax environment, this is still a GVSU space. I expect you to treat each other with respect. Bullying, harassment, etc. will not be tolerated.

Resources

Student Accessibility Resources

Grand Valley State University strives to provide an inclusive environment across campus that is accessible to all individuals with a diverse range of abilities. As your instructor, it is my objective to facilitate opportunities within all class activities and programs because your success is important to me. If you are encountering difficulties that are interrupting your learning experience please feel free to make those known to me as soon as possible, as early planning is essential.

If you feel that you need accommodations in this course, you must present a memo to me from Accessibility Resources, indicating the existence of a disability and the approved accommodations. Please reach out ASAP via email. Accommodations are not retroactive. If you have not already done so, please contact the Accessibility Resources office (215 CON) by calling (616) 331-2490 or by email to access@gvsu.edu. **Please note that I cannot provide accommodations based upon disability until I have received a copy of the Accessibility Resources issued memo.** Furthermore, if you have a disability and think you will need assistance evacuating this classroom and/or building in an emergency, please make me aware so that the university and I can develop a plan to assist you. All discussions will remain confidential.

Note: Student Accessibility Resources (SAR) was formerly called Disability Support Resources (DSR), just in case you were unaware of the name change. Their website is now: <https://www.gvsu.edu/accessibility/>

Policies

This course is subject to the GVSU policies listed at

<http://www.gvsu.edu/coursepolicies/>.

Additionally, I, as the instructor, reserve the right to modify the course policies and schedule (including due dates) over the course of the semester.

Collaboration Policy

I understand that in the real world you will rarely be asked to complete a task without discussing it with colleagues or accessing the internet. As such, I aim to balance realism and honing collaborative skills with ensuring that every student has a solid understanding of the course material.

Collaboration is encouraged for check-ins – These exercises are designed for you to gain hands-on experience, and working with those around you is not only allowed, but explicitly encouraged. However, the work you submit should fundamentally be your own. Especially for subjective questions – I really want to know what concepts you feel comfortable with and what parts you're struggling with!

Projects are to be completed individually - I want to ensure everyone understands the material, and that means that your projects need to be *your* code. However, high-level discussions are valid. Questions such as "What was the name of the command for running regex? (grep)" and "Did you have to allocate multiple chunks of memory?" are fine. Questions such as "What does the bash line look like for trimming all digits from a string?" are not. Generally, do not share your code or look at another student's code. Please also see the plagiarism and AI policies below. We will discuss this more in class, but if you have questions, please ask me!

Exams are NOT collaborative. We will discuss allowed exam materials later in the course.

Late Work Policy and Extension Days

All assignments will be posted with a due date and time. Any work submitted after the due date is late.

However, I realize that life happens. Sometimes getting work in by the due date is not feasible, for a variety of reasons. Due dates are still useful, as it prevents everyone from waiting until the semester to submit their work. As such, I include three policies to try and ease these burdens:

1. **Late Work (Projects and check-ins):** The maximum grade you can receive on a given assignment decreases by 10% per day after the due date. (e.g., at 2 days late, you can receive a maximum of $100\% - 20\% = 80\%$ on that project).

This limit caps at 3 days (70%) – I have to get grades back, and you will need to move onto the next assignment eventually!

2. **Extension days (Projects):** I give you **five (5)** project extension days for the whole semester.

These extension days can be used to extend the deadline of a project, no questions asked.

Usage details:

- a. You can use all five days on one project, or spread them across multiple projects.
 - b. These extensions must be requested on/before the due date. If you have extended your due date, you may request additional extensions before that new due date.
 - c. To request an extension, please use this Google form:
<https://forms.gle/SMaaTfXE6jqf96jj7>
 - d. You are certainly not required to use them, but there is no benefit for *not* using them, either (No bonus for saved extension days!)
 - e. Depending on when the last project is due, you may not be able to use all five extension days + late days (I have to get grades in!). I will make this very clear when the final project is assigned.
3. **Special circumstances:** If something major is going on in your life and the support of the two above policies isn't enough, please don't hesitate to reach out! I do not want you to fall behind on assignments because of a death in the family or illness, and GVSU has policies and processes to help!

Academic Honesty and Plagiarism

All students are expected to adhere to the academic honesty standards set forth by Grand Valley State University. In addition, students in this course are expected to adhere to the academic honesty guidelines as set forth by the College of Computing. Details can be found at <https://www.gvsu.edu/computing/academic-honesty-93>

Plagiarism is not tolerated and will result in a grade of 0 for the assignment in question and possibly a reduction in final letter grade. Code or ideas taken from another source must be cited. Violations will be turned in to the Dean of Students and handled per university policy. **You should be able to explain all the code that you turn in.**

Artificial Intelligence Policy

I understand that AI models are becoming increasingly useful as learning tools. As your professor, my goal (and hopefully yours too!) is for you to learn the content. Over-reliance on AI runs the risk of offloading your learning to the technology, completely negating the benefits of taking the course.

Previously, I have taught this course with a zero-AI policy. Some students struggle with it, but my exam scores were good and I was confident students were learning the material.

This semester you are allowed to use AI on programming assignments, but you must disclose it. You will be asked to explain why you used it and you must upload the transcript of your conversation. While this course is not writing-heavy, AI may not be used to generate any writing because I want to hear *your* thoughts.

I still urge you to be *very* careful in using AI in this course. Asking a large language model for clarification on a topic can be useful, and using a model as a brainstorming partner for how to conceptually breakdown a problem can be beneficial. **However, you should not be offloading your thinking to the AI model; you are here to learn!** In my opinion, critical thinking and problem solving are the *most* important skills you're honing in a computing degree, make sure you're honing *your* skills!

As stated above, you are still expected to be able to explain any code you turn in. I reserve the right to ask you questions about your work. If you cannot explain what you turned in, points may be deducted.

Religious observance

Many University students, staff, and faculty observe religious traditions from a variety of religions. The [Religious Inclusion Policy](#) acknowledges the right of students, staff, and faculty to engage in religious observances. If you need special accommodations to observe a religious holiday, **make arrangements in advance with the instructor.**

Important Dates

- For more information see the [GVSU academic calendar](#)
- 100% Tuition Refund Deadline - Friday, September 4th by 5:00 pm
- Labor Day Recess (No Class) – September 6th-7th
- 75% Tuition Refund Deadline - Friday, September 25th by 5:00 pm
- Fall break (No Class) – October 25th-27th
- Withdraw with a “W” deadline - Friday, November 13th by 5:00 pm
- Final exam week: December 12th-19th
 - Section 03 (12:00 pm) - Wednesday, December 16th, 12:00 pm - 1:50 pm
 - Section 04 (4:30 pm) – Wednesday, December 16th, 4:00 pm-5:50 pm

Tentative Schedule

This schedule is subject to (and very likely to) change as the semester progresses.

Week	Date	Topic(s)
1	2026-08-31	ssh, navigating linux, viewing files, remote file transfer
2	2026-09-07	Labor Day Recess (no class on Monday 09/06) ssh keys, history commands, IO redirection, misc. utilities
3	2026-09-14	Compression, diff, piping, vim, grep
4	2026-09-21	git
5	2026-09-28	Permissions, processes, bash scripting
6	2026-10-05	More bash scripting, regex
7	2026-10-12	sed, (maybe) gawk
8	2026-10-19	Midterm exam
9	2026-10-26	Fall Break October 25-27 Intro to C
10	2026-11-02	Memory allocation in C, pointers
11	2026-11-09	C arrays, input, and functions
12	2026-11-16	C header files, 2d arrays, structs
13	2026-11-23	Thanksgiving Recess 25-29 Tools for compiling and debugging (make, gdb, valgrind)
14	2026-11-30	C string and memory functions
15	2026-12-07	C command line args, enums/unions, file IO, libraries
16	2026-12-14	Final exam week